MIL-D-48111A (PA)
01 April 1975
Superseding
MIL-D-48111 (MU)
01 August 1972

MILITARY SPECIFICATION

DETONATOR, STAB, M94
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

*This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts and the loading, assembling and packing for one type of detonator designated as Detonator, Stab, M94.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

MILITARY

MIL-F-398 - RDX
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1375

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MIL-STD-650 - Explosives; Sampling, Inspection and Testing
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

PRODUCT AND PACKING DRAWINGS

9258184 - Box, Packing, Ammunition
9258183 - Box, Set-up, Packing, Ammunition, Outer for Small Explosive Components
9258182 - Box, Set-up, Packing, Ammunition, Inner for Small Explosive Components
9236470 - Detonator, Stab, M94

INSPECTION EQUIPMENT DRAWINGS

8816367 - Holder, firing pin
8842964 - Ball, steel
9221771 - Sleeve
9221772 - Disc, steel
9221773 - Firing pin
9221774 - Assembly, firing
9221775 - Bushing

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation E300 - Industrial Chemicals, Recommended Practice for Sampling

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn. 19103).

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3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Assembly. - The detonator assembly shall comply with all requirements specified on Drawing (dwg.) 9236470 and with all requirements specified in applicable specifications.

3.3 Moisture content

3.3.1 Primer mix NOL #130. - The moisture content at the loading station at the time of loading the detonators shall not exceed 0.30 percent.

3.3.2 Lead azide. - The moisture content at the loading station at the time of loading the detonators shall not exceed 0.50 percent.

3.3.3 Cyclonite (RDX). - The moisture content at the loading station at the time of loading the detonators shall not exceed 0.50 percent.

3.4 Functioning. - The detonator assembly shall not fail to fire on the first attempt and shall comply with the following requirements (See 6.5).

3.4.1 Output. - The detonator assembly shall function when an energy of one and one-half (1 1/2) inch ounces maximum (max.) is applied, and shall cause a 0.0055 inch minimum (min.) deep dent in a steel block of Rockwell C scale 48-51 hardness.

3.4.2 Waterproofness. - The detonator assembly shall comply with the requirements of 3.4.1 after being conditioned for forty-eight (48) hours min. at a two (2) to three (3) inch depth of water maintained at seventy (70) plus or minus two (2) degrees Fahrenhiet (F.).

3.5 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion

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products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. (Required packing shall be dry).

3.6 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

*4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a first article sample (see 6.1b) as designated by the Contracting Officer. The first article sample shall consist of 50 complete sets of parts and assemblies (i.e. 50 each of every component and assembly.) The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

*4.3.2 Inspections to be performed. - See MIL-A-48078 and paras 4.4.2.1 thru 4.4.2.7, and testing per para. 4.4.3.

*4.3.3 Rejection. - (See MIL-A-48078.)

4.4 Quality conformance inspection

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

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a. Metal parts of one lot interfix number from one manufacturer.

b. Primer mixture NOL #130 from one batch or blending of batches which results in homogeneity of all primer mixture used in loading the entire primer lot.

c. Lead azide of the same type from not more than two (2) lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

d. RDX from not more than one lot from one manufacturer.

4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48111A (PA)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.1	Cup, detonator	1	1	9236471
				NEXT HIGHER ASSEMBLY: 9236470

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Thickness of coined bottom		0.40%	3.2	Gage
Minor					
201	Length		0.65%	3.2	Gage
202	Outside diameter, maximum (max.)		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Disc, detonator closing				9236472
					NEXT HIGHER ASSEMBLY 9236470
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A&B	None defined				
Minor					
201	Diameter, minimum (min.)		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.5	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Assembly		9236470
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup split or cracked		100%	3.2	Visual
Major B					
131	Total length		0.40%	3.2	Gage
132	Outside diameter, max.		0.40%	3.2	Gage
133	Assembly distorted or damaged to extent that function may be impaired		0.40%	3.2	Visual
134	Lacquer seal missing or color incorrect		0.40%	3.2	Visual
Minor					
201	Crimp not full 360 degrees		0.65%	3.2	Visual
202	Disc wrinkled or folded		0.65%	3.2	Visual
203	Evidence of poor workmanship other than critical above		0.65%	3.5	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9258182
4.4.2.4	Box, set-up, inner, after loading and prior to closing slide cover				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Filler and cushion assembly missing		100%	3.2	Visual
2	Filler and cushion assembly improperly assembled		100%	3.2	Visual
3	Stiffner missing		100%	3.2	Visual
4	Slide cover assembly reversed		100%	3.2	Visual
Major B					
131	Release paper missing		0.40%	3.2	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Box, set-up, outer, with inner set-up boxes assembled, and prior to assembling top	1 OF 1	9258183
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Red surfaces on inner set-up boxes do not alternate on same side of outer set-up boxes		100%	3.2	Visual
2	Stiffner not facing one direction		100%	3.2	Visual
Major A&B	None defined				
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual
203	Box improperly taped		0.65%	3.2	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.6	Sealed Packed Barrier Bag, Prior to Packing in Box				9258183
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Waterproofness of barrier bag destroyed by rupture, puncture or separation of heat seal		0.40%	3.2	Visual
102	Evidence of crushed box		0.40%	3.2	Visual
NOTE	Package should be opened if there is evidence of crushing				
Minor					
201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.2	Visual
202	Marking of bag misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48111A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Box, packing, ammunition, prior to assembling cover and strapping				9258184
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Bags containing set-up boxes damaged		0.40%	3.2	Visual
132	Cushioning material incomplete or missing from outer set-up boxes		0.40%	3.2	Visual
133	Contents loose within box		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-48111A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Box, packing, ammunition				9258184
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Strapping missing, broken, loose, or improperly fastened		0.40%	3.2	Manual-Visual
102	Box damaged		0.40%	3.2	Visual
103	Marking missing, incomplete or illegible		0.40%	3.2	Visual
104	Wooden Box cover nailed to box		0.40%	3.2	Visual
105	Metallic Seal missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

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4.4.3 Testing

4.4.3.1 Chemical sampling. - A representative sample shall be randomly selected for the chemical tests specified in 4.4.3. The procedure described in ASTM E300 should be used to withdraw the samples. The chemical tests in 4.5 shall be performed using prescribed analytical procedure for replicate determination given in standard analytical textbooks.

4.4.3.2 Moisture content (see Table I) - Major A defect

TABLE I

Material
Primer mix NOL #130
Lead azide
RDX

The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification one sample of each material from each eight hours production of detonators. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement, and loading has not yet begun, that quantity of material represented by the sample shall be rejected. If detonators have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all detonators loaded with the non-conforming material shall be rejected. The tests shall be performed as specified in 4.5.1.1, 4.5.1.2 and 4.5.1.3, as applicable, using equipment in accordance with 4.4.4.

4.4.3.3 Functioning (see 3.4) - Major A defects

4.4.3.3.1 Output. - (see 3.4.1) - Major A defect - A sample of 315 detonators shall be selected at random from each lot. If 3 or more detonators fail to comply with the requirements specified, the lot shall be rejected. If one or two defectives are found, a second sample of 315 detonators shall be selected at random and tested. If in the combined first and second sample a total of 4 or more defectives are found the lot shall be rejected. The test shall be performed as specified in 4.5.2.1 using equipment in accordance with 4.4.4.

4.4.3.3.2 Waterproofness. - (see 3.4.2) - Major A defect - A sample of 25 detonators shall be selected from each lot. If

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one or more detonators fail to comply with the requirements specified, the lot shall be rejected. The test shall be performed as specified in 4.5.2.2 using equipment in accordance with 4.4.4.

4.4.3.4 Composition of primer mixture NOL #130 (see dwg. 9236470) - Major A defect. - The composition of each batch or blend of batches of primer mixture shall be determined at the time of loading. A sample of sufficient size to perform the test specified in 4.5.3 shall be selected. If the composition of the sample fails to comply with the requirements, the quantity of primer mixture represented by the sample shall be rejected. Any detonators loaded with the non-conforming mixture shall also be rejected.

4.4.4 Inspection Equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and section 6.2 herein.

4.5 Test Methods and Procedures

4.5.1 Moisture content

4.5.1.1 Primer mixture NOL #130. - Transfer a known weight of between 0.4 to 0.7 gram of sample to a tared weighing dish approximately 55 millimeters (mm) in diameter and 30 mm in height. Place the dish and contents in an oven maintained at 55 to 65 degrees Centigrade (C.) or in a desiccator containing anhydrous calcium chloride until constant weight is obtained. Calculate the loss in weight as the percentage of moisture.

4.5.1.2 Lead azide. - The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two (2) gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene to form a close fit.

4.5.1.3 RDX. - The moisture content of the RDX shall be determined in accordance with MIL-STD-650 method 101.5, except that the sample shall be heated for one (1) hour.

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4.5.2 Functioning

4.5.2.1 Sensitivity and output. - The detonator assembly shall be placed in the test fixture so that the firing pin will strike the sensitive end of the detonator. A steel ball weighing 0.50 plus or minus 0.01 ounce shall be dropped from a height of three (3) inches max, on the top of the firing pin. The drop height shall be measured as the distance between the top of the firing pin and the bottom of the ball. The dent depth in the block shall be measured to determine compliance with the requirement. Method of inspection shall be in accordance with dwgs. 8816367, 8842964, 9221771, 9221772, 9221773, 9221774, and 9221775.

4.5.2.2 Waterproofness. - The detonator assemblies shall be conditioned for forty-eight (48) hours min. at a two (2) to three (3) inch depth of water maintained at seventy (70) plus or minus two (2) degrees Fahrenheit (F). Immediately after conditioning, the detonators shall be tested as specified in 4.5.2.1. The functioning ability and dent depth shall be observed to determine compliance with the requirements.

4.5.3 Composition of primer mixture NOL #130. - The composition of the primer mixture shall be determined as follows:

4.5.3.1 Barium nitrate. - In accurately weighed portion of approximately 0.5 gram of primer mixture shall be transferred to a tared medium-porosity, fritted glass crucible of 20 ml. capacity. Three ml, of lead azide saturated distilled water shall be added at a temperature of 50 degrees C., plus or minus two degrees C., agitated for one minute and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five times, making a total of six, 3-ml. extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, dried in an oven maintained at 60 degrees C. plus or minus 5 degrees C. for thirty (30) minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barium nitrate, percent} = \frac{(A-B)100}{W(1-.01M)}$$

where:

A = weight of crucible and sample.

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B = weight of the crucible and sample after extractions with lead azide saturated water.
M = moisture content of sample obtained from 4.5.1.1 in percent
W = weight of sample

4.5.3.2 Lead styphnate

4.5.3.2.1 Basic lead styphnate (when applicable). - The residue obtained in the determination of barium nitrate shall be extracted with six 5-ml. portions of saturated ammonium acetate solution at about 25 degrees C., (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250 ml. calibrated volumetric flask the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2 ml. aliquot of this solution shall be transferred to a 50 ml. calibrated volumetric flask and diluted to the mark with distilled water. A portion of this solution shall be transferred to a corex glass spectrophotometric cell having a width of 1 centimeter (cm.) and its absorbance determined at a wave length of 410 millimicrons. As reference, a similar cell containing ammonium acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the absorbance obtained shall be corrected for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbance of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the same shall be calculated on a dry basis as follows:

$$\text{Basic lead styphnate, percent} = \frac{F(A-B)}{W(1.01M)D}$$

where:
A = absorbance of basic lead styphnate solution.
B = absorbance of ammonium acetate solution.
D = width of corex cell, cm.
M = moisture content of sample obtained from 4.5.1.1 in percent

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W = weight of sample.

F = spectrophotometric factor. To determine the spectrophometric factor (F) weigh a 0.5 gm sample of pure lead styphnate. Transfer the lead styphnate quantitatively with 50 to 100 ml of distilled water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution in the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging in concentrations from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

4.5.3.2.2 Determination of normal lead styphnate (when applicable). - Extract the residue obtained in the barium nitrate determination with sufficient 5 ml portions of ammonium acetate solution (saturated) at about 25°C, until the washings are colorless. Agitate each portion for about 1 minute, then allow each portions to remain in contact with the residue for one additional minute. Finally wash the residue with water, saturated with antimony sulfide, until the washings come through clear. Reserve the crucible and residue for the tetracene determination. Transfer the filtrate and washings quantitatively to a 250 ml calibrated volumetric flask, washing the beaker thoroughly with distilled water and dilute to the mark with distilled water. Using a calibrated pipette, transfer a 2 ml aliquot of this solution to a second 100 ml calibrated volumetric flash and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell having a width of 1 cm and determine its optical density at a wave length of 410 millimicrons. Use as reference, a similar cell containing ammonium acetate solution of the same concentration as the sample in the final dilution. Since the cell which holds the normal lead styphnate is not identical with the reference cell, correct the absorbance obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with ammonium acetate solution, and measure at a wave length of 410 millimicrons the absorbance of the cell which originally contained the normal lead styphnate in the sample on a dry basis as follows:

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$$\text{Normal lead styphnate, percent} = \frac{F(A-B)}{WD}$$

where:

A = absorbance of normal lead styphnate solution.
B = absorbance of ammonium acetate solution.
D = width of corex cell, cm.
W = weight of the dried sample.
F = spectrophometric factor. To determine the spectrophometric factor (F) weight a 0.5 gm sample of pure lead styphnate. Transfer the lead styphnate quantitatively with 50 to 100 ml of distilled water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution in the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

4.5.3.3 Tetracene. - The wet residue obtained in the determination on lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60 degrees C. plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125 ml. beaker with a stream of water, 25 ml. of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water, and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60 degrees C. plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

$$\text{Tetracene, percent} = \frac{(C-D)\ 100}{W(1-.01M)}$$

where:

C = weight of crucible and residue after extraction with saturated ammonium acetate solution.
D = weight of crucible and residue after treating with hot water.

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W = weight of the sample.
M = moisture content of sample obtained from 4.5.1.1 in percent.

4.5.3.4 Antimony sulfide. - The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)}{W(1-.01M)}$$

where:
D = weight of crucible and sample after treating with hot water.
E = weight of crucible.
W = weight of sample.
M = moisture content of sample obtained in 4.5.1.1 in percent.

4.5.3.5 Lead azide. - The lead azide content in the sample on a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, lead styphnate, tetracene and antimony sulfide.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C. - The detonators shall be preserved and packaged in accordance with dwgs. 9258182 and 9258183.

*5.1.2 Level C. - The detonator parts shall be packaged in accordance with MIL-STD-1169 requirements.

5.2 Packing

5.2.1 Level C. - The detonators shall be packed in accordance with 9258184.

*5.2.2 Level C. - The detonator parts shall be packed in accordance with MIL-STD-1169 requirements.

*5.3 Marking. - Detonator containers and bags shall be marked in accordance with instructions contained in dwgs. 9258182, 9258183 and 9258184. Detonator parts containers shall be marked in accordance with MIL-STD-1169 requirements.

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6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Provisions of MIL-A-48078.

6.2 Submission of In Spection Equipment Designs for Approval. (See MIL-A-48078). Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T Dover, New Jersey 07801.

6.3 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M Dover, New Jersey 07801

*6.4 Check test for possible deterioration. - The following test is designed to prevent detonators that have deteriorated from being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Detonator assemblies shall be selected from each lot for this test.

*6.5 Output. - Sampling shall be in accordance with MIL-STD-105 using Code letter P with an AQL of 0.15 percent. The test shall be performed as specified in 4.5.2.1, using equipment as specified in 4.4.4.

6.7 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.

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Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1375-A127

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-A
Dover, New Jersey 07801

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